Figure 16: OpenDiffraction Benchmark Tasks (T1 – T12) Specification

Task ID	Task Name	Problem Type	Input Modalities	Target Modalities	Primary Metric
T1	Phase Retrieval	Inverse Problem	intensity_measured, mask	target_dpr, object_amplitude	`ALIGNED_MSE`
T2	Diffraction Pattern Denoising	Image Restoration	intensity_measured	intensity_expected	`PSNR`
T3	Detector Inpainting	Spatial Inpainting	intensity_measured, mask	intensity_expected	`MASKED_PSNR`
T4	Hit Finding and Classification	Classification	intensity_measured	hit_status	`BALANCED_ACCURACY`
T5	Detector Defect Segmentation	Semantic Segmentation	intensity_measured	defect_mask	`IOU`
T6	Specimen Morphology Classification	Classification	intensity_measured	morphology_class	`TOP1_ACCURACY`
T7	Physical Parameter Estimation	Regression	intensity_measured	photon_budget, coherence_length	`RELATIVE_MAE`
T8	Out-of-Distribution Anomaly Detection	Anomaly Detection	intensity_measured	is_ood	`AUROC`
T9	Domain Generalization	Domain Adaptation	intensity_measured, mask	target_dpr	`TRANSFER_DEGRADATION_RATIO`
T10	Self-Supervised Representation Learning	Self-Supervised Learning	intensity_measured	representation_vector	`LINEAR_PROBE_ACCURACY`
T11	Uncertainty Quantification	Probabilistic Inverse Problem	intensity_measured, mask	target_dpr, pixelwise_variance	`EXPECTED_CALIBRATION_ERROR`
T12	Forward-Inverse Physics Consistency	Constraint Verification	reconstructed_object, intensity_measured, mask	consistency_error	`R_FACTOR`